

Pooling and spatial information

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Convolution as a local operation

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learned representations

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Regularisation and early stopping

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Prove that location can be lost for Pooling and spatial information. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Pooling and spatial information. Proposed explanation: “Max pooling turns negative numbers into zero”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Pool a new map independently

Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Calculate both summaries

For the lesson window starting at [2, 2], list the four values, sum, maximum and mean. State the complete pooled output shape.

My reasoning and evidence

6. W2 Prove that location can be lost

Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

My reasoning and evidence

7. W3 Audit an omitted border

Apply the lesson complete-window rule to $[[1, 2, 99], [3, 4, 99], [99, 99, 99]]$. Give the output shape and both outputs. Explain why neither output includes 99.

My reasoning and evidence

8. W4 Pool a new map independently

Use rows $[-4, -1, 2, 6]$, $[-3, -2, 4, 0]$, $[5, 1, -8, -2]$, $[3, 7, -4, -6]$. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

My reasoning and evidence

9. W5 Check the movement boundary

A 2 by 4 map is zero except for 8 at row 0 column 1. Move the 8 to column 2 and set its old position to zero. Give the max and mean maps before and after. What claim does this refute?

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled pooling and spatial information evidence sheet to Image-pattern Network Investigation. Show how you can compute max and mean pooling under an explicit window rule. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Pooling and spatial information

Pool a small map and show the detail it cannot retain.

Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Two adjacent pooling windows

	left0	left1	right0	right1
row0	1	4	2	0
row1	3	2	5	1

Read a cell using both its row and column labels.

Pooling and spatial information

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Left window

Values 1, 4, 3, 2: max 4, mean 2.5.



2. Right window

Values 2, 0, 5, 1: max 5, mean 2.



3. Compare summaries

Max map $[[4, 5]]$, mean map $[[2.5, 2]]$, both shape 1 x 2.

CHANGE ONE THING

Swap 1 and 4 within the left window.

Explain the changed result using the matching lesson.

Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Pool a small map and show the detail it cannot retain.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Max map $\begin{bmatrix} 4 & 5 \end{bmatrix}$, mean map $\begin{bmatrix} 2.5 & 2 \end{bmatrix}$, both shape 1×2 .

1. Left window
2. Compare summaries
3. Right window

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare summaries Max map $\begin{bmatrix} 4 & 5 \end{bmatrix}$, mean map $\begin{bmatrix} 2.5 & 2 \end{bmatrix}$, both shape 1×2 . Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared. Swapping values inside one window leaves both summaries unchanged here, so the pooled output cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the output shape.

Practice 2: explain the deciding evidence

Changed-case practice: Swap 1 and 4 within the left window. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Both pooled maps remain unchanged.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Both pooled maps remain unchanged. Max and mean ignore the positions of values inside the same window.

Repair a missing building block

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an input through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from prediction error. Short practice: Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
